

Testing Report — ProBuild Supplies Ltd

Operational Business Process Automation

UFCFAF-30-3 DISP Group 4 | Submission Date: 23 April 2026

System under test: Camunda 8 Operational BPMN — ProBuild Supplies Ltd **Process model:** Operational_Business_Process_Model.bpmn (4 pools) **Engine:** Camunda 8.8.14 (self-managed, c8run) **Stack:** Spring Boot 3.4.5, Java 21, H2 in-memory, JPA/Hibernate **Repository:** DISP-Group4 / Probuild-Project

Version	Date	Author	Notes
0.1	May 2026	DISP G4	Initial plan
1.0	May 2026	DISP G4	Pre-execution shell, 41 test cases
1.1	TBC	DISP G4	Post-execution: actual results added

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1. Introduction

1.1 Purpose

This report documents the test strategy, test cases, execution evidence and defect analysis for the ProBuild Supplies Ltd automated business process. It serves as both an executable test plan and the final summative testing report for the portfolio submission.

The system under test is a Camunda 8 process automation covering four pools (Customer, FinTrust UK, ProBuild LTD, FixPro Ltd), backed by a Spring Boot job worker application with an H2 persistence layer and twelve Camunda Forms.

1.2 Scope

In scope

- All four pools and their automated service tasks
- All twelve Camunda user task forms including conditional visibility and field validation
- Message start events and intermediate catch events for cross-pool correlation
- Database persistence: tool stock, bookings, trade card members
- Business rules: Trade Card discount, finance eligibility, repair authorisation threshold
- REST API endpoints
- End-to-end happy-path and exception-path scenarios

Out of scope

- Load and performance testing
- Security and penetration testing
- Real external API integration with FinTrust/FixPro
- Loyalty points accrual and tiered discount bands (known limitation — see §8)
- Internationalisation and accessibility

1.3 Definitions

Term	Meaning
BPMN	Business Process Model and Notation 2.0
Camunda	Camunda 8 — the process automation platform
c8run	Self-managed bundled Camunda 8 distribution
Operate	Camunda monitoring UI for process instances

Tasklist	Camunda UI for human task completion
Job worker	Spring component annotated <code>@JobWorker</code> that activates and completes jobs
FEEL	Friendly Enough Expression Language — used in gateway expressions
TC	Trade Card (context-dependent); Test Case when prefixed with TC-

2. Test Strategy

2.1 Approach

A risk-based manual integration testing approach has been adopted. End-to-end flow execution through Tasklist and Operate, supplemented by direct database and API inspection, provides the primary evidence. Where execution is costly or low-value, inspection-based verification of the BPMN XML, form JSON, and worker source code supplements execution evidence.

2.2 Test Types

Type	Purpose	Tools
Integration (end-to-end)	Verify complete process flows across all pools	Tasklist, Operate, Spring logs
Business logic	Verify worker calculations and conditional logic	Operate variables, H2, logs
Form validation	Verify field constraints, regex patterns, conditional show/hide	Tasklist, form JSON inspection
Database persistence	Verify state transitions are correctly persisted	H2 console, REST API
Message correlation	Verify cross-pool messages reach the correct instance	Operate, worker logs
Negative / error path	Verify behaviour on invalid or boundary inputs	Tasklist, Operate, logs

Process integrity	Verify no orphaned tokens, dead-end paths, or unreachable activities	BPMN inspection, Operate
Regression	Re-run previously failed tests after fixes	Re-run of relevant TC-entries

2.3 Priority Classification

Priority	Definition
P1 — Critical	Core business rule or flow; failure invalidates the demonstration
P2 — High	Important business rule or persistence boundary
P3 — Medium	User-facing constraints and conditional UI
P4 — Low	Edge cases and inspection-based checks

2.4 Test Environment

Setting	Value
OS	macOS (darwin / arm64)
Java	21
Spring Boot	3.4.5
Camunda 8	8.8.14 (c8run)
Database	H2 in-memory — jdbc:h2:mem:probuilddb;DB_CLOSE_DELAY=-1
App port	9090
Camunda gRPC / REST	127.0.0.1:26500 / 127.0.0.1:8080

2.5 Test Data

DataSeeder populates a deterministic baseline on every application start:

Trade card seed data

Card number	Holder	Active
TC-1000000001	James Builder	true
TC-1000000002	Sarah Contractor	true
TC-1000000003	Mike Tradesman	true
TC-1000000004	Lisa Plumber	true
TC-1000000005	Dave Electrician	true

Selected tool data

Tool	Purchase £	Hire/day £	Deposit £	Stock
Cordless Drill	80.00	12.00	30.00	15
Hammer Drill (SDS)	110.00	18.00	40.00	8
Concrete Breaker	350.00	60.00	120.00	3
Cement Mixer	120.00	25.00	50.00	5
Strimmer	65.00	10.00	25.00	10
Scaffolding	200.00	40.00	100.00	3

2.6 Entry and Exit Criteria

Entry criteria (must be true before testing begins)

- c8run running; Operate, Tasklist and Zeebe gateway reachable
- Spring Boot starts cleanly; all 28 job workers registered in console output
- DataSeeder confirms 27 tools and 5 trade cards seeded
- BPMN process deployed and visible in Operate
- GET /api/tools returns 27 entries

Exit criteria (testing complete when)

- Every P1 and P2 test has an Actual result and Pass/Fail/Partial status
 - Every P3 test executed or formally deferred with justification
 - Every defect logged in §6 with severity, root cause and resolution status
 - Coverage analysis in §7 completed
 - Limitations accurately reflected in §8
-

3. Traceability Matrix

3.1 BPMN Element Coverage

BPMN Element	Type	Worker / Form	Covering Tests
In Store or Online?	User task	choose-shopping-method	TC-P01, TC-P02, TC-H01
Check Online Availability	Service task	checkOnlineAvailability	TC-P01, TC-H01
Check In Store Availability	Service task	checkInStoreAvailability	TC-I02
Select Tools	User task	select-tool	TC-P01, TC-P03, TC-H01
Collect Customer Information	User task	collect-customer-information	TC-P01, TC-V01, V02, V04, V05
Calculate Price	Service task	calculatePrice	TC-P03, TC-P04, TC-B01, TC-B02
Register Customer as Member	Service task	registerMember	TC-D03
Get Payment Details	User task	get-payment-details	TC-P01, TC-V03, V06, TC-F02
Send Credit Application	Send task	sendCreditApplication	TC-F01, F02, F04

Process Credit Application	Service task	processCreditApplication	TC-F01–F04, TC-B03
Set up repayment schedule	Service task	setupRepaymentSchedule	TC-F01
Send Repayment Plan	Send task	sendRepaymentPlan	TC-F01
Send Finance Decision	Send task	sendFinanceDecision	TC-F01, F02
Pay ProBuild agreed amount	User task	pay-probuild-amount	TC-F01
Process Payment	Service task	processPayment	TC-P01
Choose delivery method	User task	choose-delivery-method	TC-P01, P02
Select a delivery Time	User task	select-delivery-time	TC-P01
Schedule Delivery	User task	schedule-delivery	TC-H01
Update System when delivered	User task	update-system-delivered	TC-P01
Notify Pickup Ready	Send task	notifyPickupReady	TC-P02
Notify Customer (delivery)	Send task	notifyCustomer	TC-P01
Send Hire Request	Send task	sendHireRequest	TC-H01, TC-M01
Duration of Hire / Dates	User task	duration-of-hire	TC-H01, V01, V02

Check Tool Availability	Service task	checkToolAvailability	TC-H01, H02
Confirm Hire Booking	Service task	confirmHireBooking	TC-H01, D01, D04
Send Booking Confirmed	Send task	sendBookingConfirmed	TC-H01, TC-M01
Send Tool Unavailable	Send task	sendToolUnavailable	TC-H02
Receive Tool Hire Information	User task	hire-summary	TC-H01
Send Purchase Detail	Send task	sendPurchaseDetail	TC-P01
Return Tool	Send task	sendReturn	TC-H01, H03–H05
Inspect Tool and handover	User task	inspect-tool-handover	TC-H03–H05, TC-N02
Send Tool for Maintenance	Send task	toolsForMaintenance	TC-H03–H05, TC-M02
Send Repair Authorisation	Send task	sendRepairAuthorisation	TC-H03, H04
Send Approval / Disapproval	Send tasks	sendApproval / sendDisapproval	TC-H03, H04, TC-N03
Send Tools Back	Send task	sendToolsBack	TC-H01, TC-M02
Mark Tool Available	Service task	markToolAvailable	TC-D02, TC-H01

3.2 Business Rule Coverage

Rule	Worker	Tests
10% Trade Card discount	calculatePrice	TC-P03
Trade card validated against DB	calculatePrice	TC-P03, P04
Finance requires total $\geq$ £100	processCreditApplication	TC-F02, B03
Finance requires total $<$ £5,000	processCreditApplication	TC-F04
Repair cost $>$ £50 flags manager auth	sendRepairAuthorisation	TC-H03, H04, B04
Stock decrements on hire booking	confirmHireBooking	TC-D01, H01
Stock restored on return	markToolAvailable	TC-D02, H01
Booking reference auto-generated BK-{millis}	confirmHireBooking	TC-H01, D04
Monthly repayment = total / period	setupRepaymentSchedule	TC-F01
Trade card regex ^TC-[0-9]{10}\$	Form	TC-V04
Card number regex ^[0-9]{4}\$	Form	TC-V03
Phone regex ^[0-9+\s-()]{10,15}\$	Form	TC-V01

3.3 Case Study Requirement Coverage

Requirement	Status	Tests
Purchase tools online or in-store	Implemented	TC-P01, P02
Stock updated in real time	Implemented	TC-D01, D02
Delivery or click-and-collect	Implemented	TC-P01, P02
Payment by cash, card or finance	Implemented	TC-V06, P01, F01

Hire tools for a defined period	Implemented	TC-H01
Tool hire takes booking and deposit	Implemented	TC-H01, D04
Tools inspected on return	Implemented	TC-H03–H05
Maintenance via FixPro Ltd	Implemented	TC-H03, H05, M02
Repairs > £50 require auth	Implemented (logged, not blocked)	TC-H04, B04
Trade Card £10/year fee	Partial — registration exists, payment not modelled	TC-D03
Finance for Trade Card holders only	Partial — not enforced in worker	TC-F03
10% loyalty discount	Partial — flat 10%, no points engine	TC-P03
Finance for purchases ≥ £100	Implemented	TC-F02, B03
Finance over 6 or 12 months	Implemented	TC-F01
Quarterly KPI review	Out of scope	—
Weekly FixPro collection cycle	Out of scope (no timer events)	—

4. Test Cases

Status key: ☐ Pass ☐ Fail ☐ Partial/Blocked

Each test block includes: ID, type, priority, steps, expected result, evidence placeholders, and space for actual results.

4.0 Pre-Flight Checks

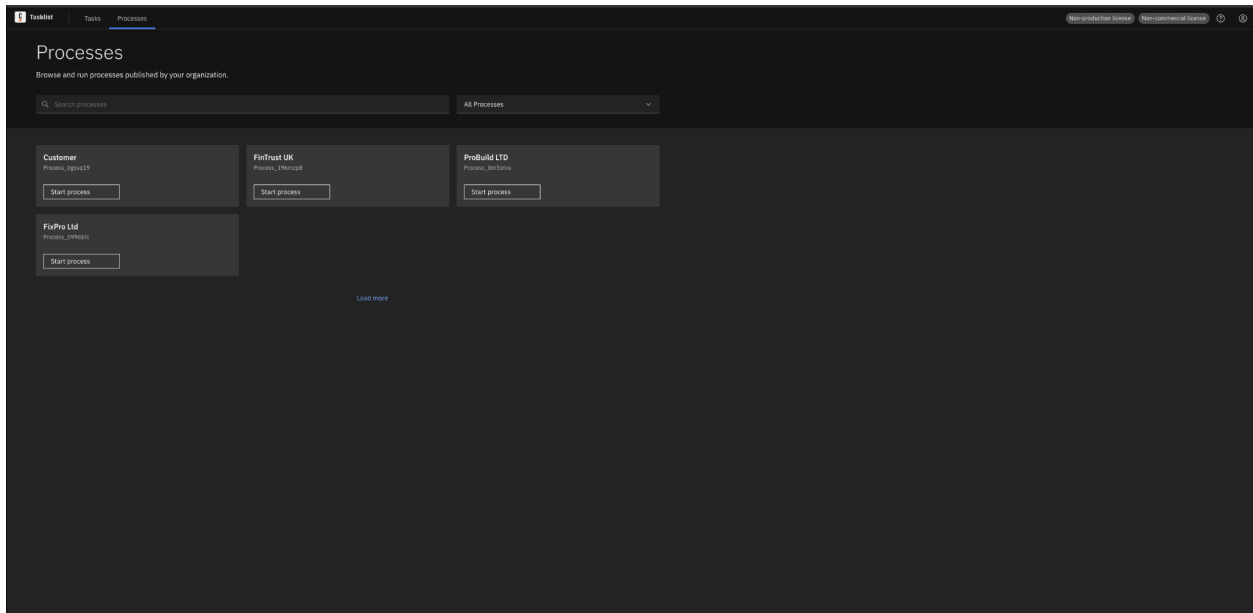
PRE-01 — Runtime components reachable Type: Smoke | Priority: P1

Steps:

1. Start c8run
2. Start Spring Boot: `./mvnw spring-boot:run`
3. Open Operate at `http://localhost:8080/operate` and Tasklist at `http://localhost:8080/tasklist`
4. Open H2 console at `http://localhost:9090/h2-console` (JDBC: `jdbc:h2:mem:probuilddb`, user: `sa`)
5. GET `http://localhost:9090/api/tools`

Expected: Operate and Tasklist render; H2 connects; `/api/tools` returns 27 entries.

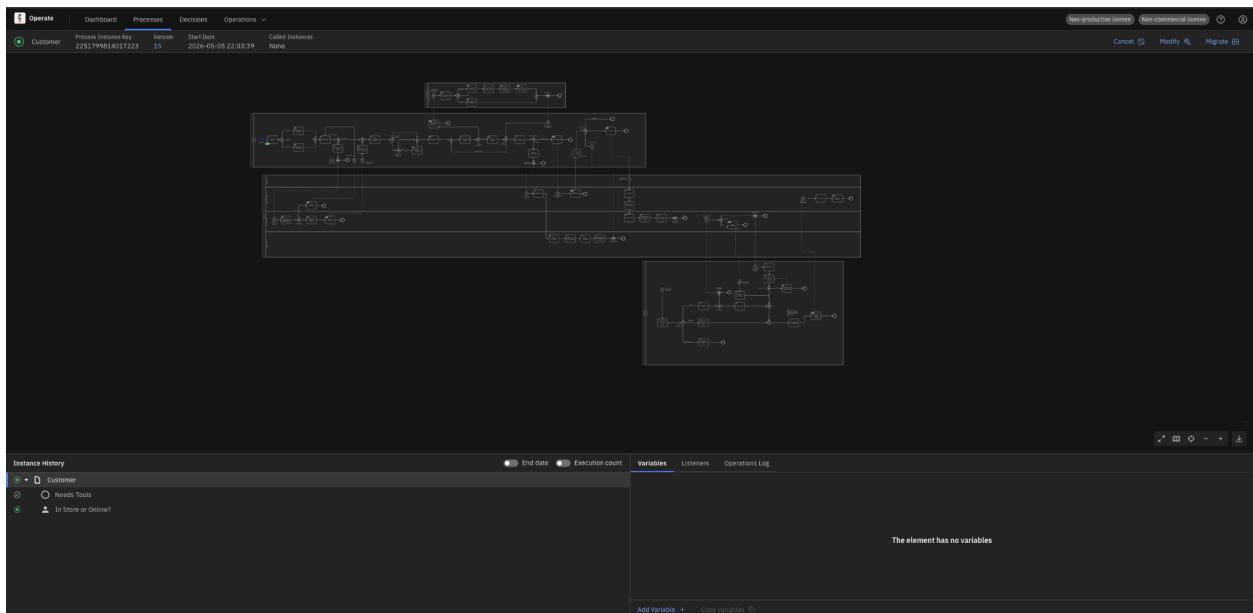
Evidence:



Operate landing

AVAILABLE_QUANTITY	DISPOST_AMOUNT	HIRE_RATE_PER_DAY	PURCHASE_PRICE	ID	DISPLAY_NAME	NAME
15	36.0	12.0	85.0	1	Cordless Drill	cordless-drill
8	44.0	14.0	110.0	2	Hammer Drill (SD)	hammer-drill
12	25.0	10.0	55.0	3	Angle Grinder	angle-grinder
8	36.0	12.0	75.0	4	Circular Saw	circular-saw
10	25.0	10.0	50.0	5	Jigsaw	jigsaw
7	28.0	12.0	70.0	6	Reciprocating Saw	reciprocating-saw
3	75.0	30.0	180.0	7	Plan Band	plan-band
7	25.0	12.0	70.0	8	Sheet Metal	sheet-metal
6	30.0	14.0	85.0	9	Wood Router	router
10	35.0	15.0	90.0	10	Paint Sprayer	paint-sprayer
10	25.0	10.0	60.0	11	Blower	blower
8	28.0	8.0	50.0	12	Leaf Blower	leaf-blower
5	90.0	35.0	140.0	13	Chainsaw	chainsaw
4	55.0	20.0	100.0	14	Petrol Lawnmower	lawnmower
3	90.0	40.0	220.0	15	Mower	mower
5	90.0	35.0	150.0	16	Concrete Mixer	concrete-mixer
8	40.0	18.0	95.0	17	Pressure Washer	pressure-washer
8	35.0	15.0	80.0	18	Tile Cutter	tile-cutter
3	180.0	60.0	200.0	19	Scaffolding	scaffolding
3	120.0	60.0	200.0	20	Concrete Breaker	concrete-breaker
4	180.0	60.0	200.0	21	Generator	generator
5	30.0	20.0	100.0	22	Demolisher	demolisher
4	85.0	30.0	180.0	23	Air Compressor	air-compressor
12	20.0	8.0	70.0	24	Extension Cord	extension-cord
8	30.0	12.0	60.0	25	Potholes Steps	potholes-steps
20	15.0	5.0	40.0	26	Wheelbarrow	wheelbarrow
8	25.0	10.0	60.0	27	Site Lighting	site-lighting

The database schema is visible confirming the application has started and DataSeeder has run.



Camunda model is operational

Actual result: All four runtime components started successfully. Operate and Tasklist rendered at localhost:8080, the H2 console connected with the seeded credentials, and GET /api/tools returned all 27 expected tool entries. No errors were observed during startup.

Status: Pass | Tester: Sam | Date: 05/05/2026 | Build: 1.0

All tools are 27 rows

jdbc:h2:mem:probuiddb

BOOKINGS

TOOLS

TRADE_CARDS

INFORMATION_SCHEMA

Users

H2 2.3.232 (2024-08-11)

Max rows: 1000

Auto complete

Off

Auto select

On

Run

Run Selected

Auto complete

Clear

SQL statement:

SELECT * FROM TRADE_CARDS

SELECT * FROM TRADE_CARDS;

ACTIVE	ID	CARD_NUMBER	HOLDER_NAME
TRUE	1	TC-1000000001	James Builder
TRUE	2	TC-1000000002	Sarah Contractor
TRUE	3	TC-1000000003	Mike Tradesman
TRUE	4	TC-1000000004	Lisa Plumber
TRUE	5	TC-1000000005	Dave Electrician
TRUE	6	TC-6014406972	Nick LeMasonry
TRUE	7	TC-6016192723	Nick LeMasonry

(7 rows, 1 ms)

Edit

Seven Trade active five initially and two more post tests as both tests "I'd like to register for a trade card membership" were selected for Nick Lemasonry.

Actual result: The H2 console queries returned a TOOLS count of 27 and a TRADE_CARDS count of 7, matching the expected seeded baseline.

Status: Pass | Tester: Sam | Date: 05/05/2026 | Build: 1.0

4.1 Purchase Flow

This section describes the process of buying, whether it is in-store or online.

This begins with selecting the buyer's preference. With online buying, the system checks the stock online first, and in-store stock on the shop floor for buyers who want to purchase in person.

When stock availability is verified, the buyer selects their purchasing method, fills up the billing section, and enters their contact information. If applicable, the buyer can enter their Trade Card number. This will allow the buyer to get an automated discount, once the system verifies the number in the database.

After this, their mode of payment is selected. This can either be cash, card, or financed. For payment of finance, the system defaults the process to the customized flow of §4.2. This is manually processed, however, cash or card payment is processed immediately.

Post payment, the buyer is asked their preference for how the order is to be received for home delivery, they now choose their delivery times. From here, the system will notify when the order is on its way. For center collection instead, the system will notify once the order is ready to be collected.

During the order collection and delivery, the payment is logged and the system maintains a purchase record until the order is officially processed.

TC-P01

1. Start new instance → In Store or Online? → Online
2. Select Tools → Cordless Drill, Purchase, qty 1
3. Collect Customer Information → fill all fields, no Trade Card
4. Get Payment Details → Card, enter 1234
5. Choose delivery method → Home Delivery
6. Select a delivery Time → any slot
7. ProBuild: Schedule Delivery → assign van and driver
8. Complete Pack/Deliver manual tasks
9. Update System when package delivered

Expected Result:

Test data:

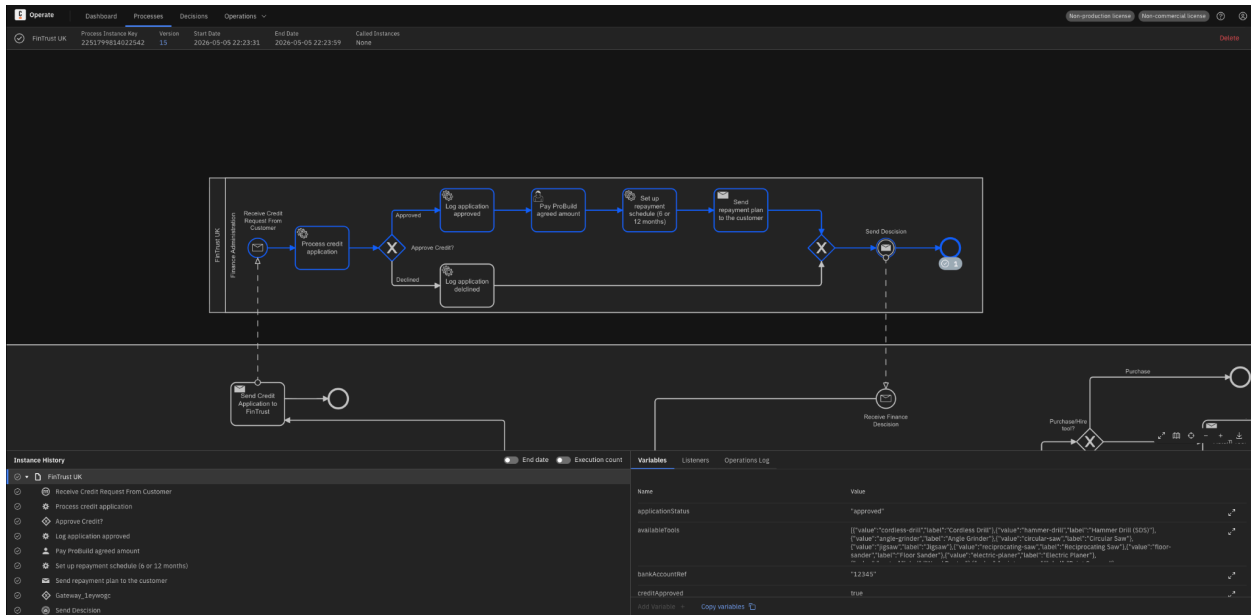
Test ID	Test Description	Pass / Fail
TC-P01	In-store purchase — item in stock, card payment succeeds	Pass
TC-P02	Online purchase — delivery scheduled and system updated on delivery	Pass
TC-P03	Purchase — item out of stock, customer notified	Pass
TC-P04	Purchase with Trade Card — loyalty discount applied	Pass

4.2 Finance Flow

This section covers the finance flow, which is triggered when a customer selects “Finance” as their payment method during checkout. The flow begins in the Customer pool after the customer provides their payment details, which prompts a Send Credit Application task to dispatch a message to the FinTrust UK pool. FinTrust’s Process Credit Application service task then

evaluates eligibility — the application is approved only if the total is between £100 and £5,000. If approved, FinTrust sets up a repayment schedule, sends the plan back to the customer, and the customer completes a Pay ProBuild Agreed Amount user task to confirm the financed portion. A Send Finance Decision message is returned to the Customer pool regardless of outcome, directing the process toward payment completion on approval or back to the payment selection screen on rejection. The screenshots below show the full approved-finance path as observed in Camunda Operate and Tasklist, including the pre-filled finance amount on the FinTrust form and the repayment schedule variables confirmed in Operate.

Test ID	Test Description	Pass / Fail
TC-F01	Finance selected — FinTrust credit check returns approved	Pass
TC-F02	Finance selected — FinTrust credit check returns declined, redirected to alt payment	Pass
TC-F03	Finance blocked — customer is not a Trade Card member	Fail
TC-F04	Repayment months field shown when finance selected, hidden for card/cash	Pass



1. Show how the process work

Tasklist | Tasks | Processors

Non-production license | Non-commercial license

Don't miss new assignments Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

Assigned to me | Unassign

Details

Creation date: 03 May 2020 - 10:21 PM

Candidates: No candidates

Priority: Medium

Due date: No due date

Collect customer information
Customer

Task | Process | History

Customer Information

Please confirm your details below. If you submitted a hire request, your name and email will have been pre-filled – just complete the remaining fields.

Personal Details

First Name*
NICK

Last Name*
LeMasonry

Email Address*
rick@nick.com

Phone Number*
1234567890
UK format – numbers, spaces, hyphens and + only.

Address*
NICK'S house

Trade Card Membership

☐ I am an existing Trade Card Member

☒ I would like to register for a Trade Card Membership (£10/year)

Auto-select first available task

Complete Task

2. Input customer information for finance payment method

Tasklist | Tasks | Processors

Non-production license | Non-commercial license

Don't miss new assignments Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

Assigned to me | Unassign

Details

Creation date: 03 May 2020 - 10:23 PM

Candidates: No candidates

Priority: Medium

Due date: No due date

Get Payment Details
Customer

Task | Process | History

Payment Details

Your order total is shown below. Select your payment method to continue.

Original Price (£)
125

Trade Card Discount (20% off)
25

UKK: Needs discount for verified Trade Card members.

Total Amount Due (£)
100.00

Payment Method*
Interest-Free Finance (Trade Card members only)

Finance Payment Details

Interest-free finance is available to Trade Card members only. Your trade card number is shown below.

Trade Card Number
TC-1034567890123
Confirmed from your customer information.

Repayment Period*
☒ 6 Months
☐ 12 Months

☒ I confirm I am applying for interest-free finance and agree to the repayment schedule

Auto-select first available task

Complete Task

3. The customer can choose 6/12 months for finance payment if they select finance payment

Tasklist

TasksProcesses

All open tasks1

Pay ProBuild agreed amount

FinTrust UK

9 hrs

Close

Medium

Pay ProBuild agreed amount

FinTrust UK

TaskProcessHistory

Don't miss new assignments

Turn on notifications

Turn on notifications in your browser to get notified when new tasks are assigned to you.

Assigned to me

Unassign

Details

Creation date

03 May 2024 - 10:23 PM

Candidates

No candidates

Priority

Medium

Due date

No Due Date

Finance Repayment Agreement

Your credit application has been approved by FinTrust UK. Please review the terms below and confirm your repayment preference.

Finance Amount Approved GBP*

100.0

Repayment Period*

6 Months

Bank Account Reference*

12345

Enter your sort code and account number (e.g. 12-34-56 / 12345678).

☒ I agree to the repayment schedule set by FinTrust UK*

Complete Task

Auto-select first available task

4. The agent will approve for the finance payment if the total amount is above £100

Payment Details

Your order total is shown below. Select your payment method to continue.

▲ Finance application was declined: Finance is only available for purchases of £100 or more. Please select an alternative payment method.

Order Summary

Original Price (£)

110

- | +

Trade Card Discount (10%) (£)

11

- | +

10% loyalty discount for verified Trade Card members.

Total Amount Due (£)*

99

- | +

Payment Method

Payment Method*

Interest-Free Finance (Trade Card members only)

x | v

Finance Payment Details

Interest-free finance is available to Trade Card members only. Your trade card number is shown below.

Trade Card Number

TC-6022805623

Confirmed from your customer information.

Repayment Period*

☒ 6 Months

☐ 12 Months

5. The system will decline any purchase or hire that lower than £100 automatically

4.3 Hire and Return Flow

This section explains the entire process of tool hiring and returning. When a tool is hired, a request that initiates a discussion between the Customer pool and ProBuild LTD is sent. ProBuild fills the request with the tool and hire dates, and the system verifies availability. If the tool is available, a booking is created, stock is decremented, and a booking reference is created and assigned. Afterwards, ProBuild sends the customer booking details for final review. If a tool is not available, the customer receives a request denial notice.

First are the screenshots for the 'happy path', showing the customer request flow and ProBuild booking fulfillment flow. The second shows the customer denial notice and the end of the request. If the customer returns a tool, a ProBuild employee inspects the tool and documents their findings. Tools with no damages are immediately added to stock. Tools with noted damages are sent to FixPro for repair. If the repair cost is > £50, the system requests a repair authorisation from the manager. After the tool is repaired and cleared of manager authorisation, it is returned to stock.

Test ID	Test Description	Pass / Fail
TC-H01	Hire flow — tool available, handover form completed, IMS updated	Pass
TC-H02	Hire flow — tool unavailable, customer informed	Pass
TC-H03	Hire duration form — valid dates accepted, hireDays calculated	Pass
TC-H04	Hire duration form — end date before start date rejected	Fail
TC-H05	Inspect tool handover checklist submitted successfully	Pass
TC-H06	Hire summary — correct deposit and daily rate per tool pricing catalogue	Pass

DELIVERY (Logistics Team)		
Test ID	Test Description	Pass / Fail
TC-D01	Schedule delivery — valid slot accepted, process advances	Pass
TC-D02	Select delivery time — time slot options displayed correctly	Pass
TC-D03	Update system on delivery — IMS updated, process ends	Pass

TOOL RETURN & FIXPRO LTD

Test ID	Test Description	Pass / Fail
TC-R01	Tool return triggers message to FixPro, new FixPro instance started	Pass
TC-R02	FixPro service complete — tool reintegrated, IMS updated	Pass
TC-R03	Repair cost > £50 triggers authorisation request to ProBuild Tool Hire Manager	Pass
TC-R04	Repair cost < £50 - FixPro proceeds with repair directly, no authorisation required	Pass

Tasklist

Tasks

Processes

Non-production testing

Non-commercial license

Don't miss new assignments

Turn on notifications

Assigned to me

Unassign

Creation date

03 May 2024 - 12:10 PM

Candidates

No candidates

Priority

Medium

Due date

No due date

All open tasks

In Store or Online?

Customer

No

Now

Medium

In Store or Online?

Customer

Task

Process

History

Welcome to ProBuild Supplies

Please select how you would like to shop today.

How would you like to browse our catalogue?

☐ Online

☒ In Store

Auto-select first available task

Complete Task

Tasklist

Tasks

Processes

Non-production testing

Non-commercial license

Don't miss new assignments

Turn on notifications

Assigned to me

Unassign

Creation date

03 May 2024 - 10:11 PM

Candidates

No candidates

Priority

Medium

Due date

No due date

Select Tools

Customer

Medium

Select Tools

Customer

Task

Process

History

Select Tools

Choose whether you would like to purchase or hire, then select the tools you need from our available stock.

Purchase or Hire?

☐ Purchase

☒ Hire

Select Tools / Material?

Open

Search

Only tools currently in stock are shown.

Quantity?

1

-

+

Auto-select first available task

Complete Task

tasklist

TasksProcesses

Non-production tasksNon-commercial tasks

⌵

All open tasks

4

Duration of Hire / Dates

Customer

🔔 No

🔊 Medium

📌 Hire

Duration of Hire / Dates

Customer

TaskProcessHistory

ⓘ Don't miss new assignments

Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

✕

Assigned to me

Unassign

Details

Creation date

05 May 2020 - 10:11 PM

Candidates

No candidates

Priority

Medium

Due date

No due date

Hire Duration & Contact Details

Enter your contact details and select your hire period. These details will be used to confirm your booking.

First Name*

Nick

Last Name*

LeMasonry

Email Address*

nick@nick.com

Phone Number*

12345678910

UK format – numbers, spaces, hyphens and + only.

Hire Period

+

06/05/2026

+

13/05/2026

Auto-select first available task

Complete Task

tasklist

TasksProcesses

Non-production tasksNon-commercial tasks

⌵

All open tasks

4

Collect customer information

Customer

🔔 No

🔊 Medium

📌 Hire

Collect customer information

Customer

TaskProcessHistory

ⓘ Don't miss new assignments

Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

✕

Assigned to me

Unassign

Details

Creation date

05 May 2020 - 10:12 PM

Candidates

No candidates

Priority

Medium

Due date

No due date

Customer Information

Please confirm your details below. If you submitted a hire request, your name and email will have been pre-filled – just complete the remaining fields.

Personal Details

First Name*

Nick

Last Name*

LeMasonry

Email Address*

nick@nick.com

Phone Number*

12345678910

UK format – numbers, spaces, hyphens and + only.

Address*

Nick's house

Trade Card Membership

☐ I am an existing Trade Card Member

☐ I would like to register for a Trade Card Membership (£10/year)

Auto-select first available task

Complete Task

Tasklist

Tasks

Processes

Non-production tasks

Non-confidential tasks

All open tasks

-1

Get Payment Details

Customer

High

Medium

Close

Get Payment Details

Customer

Task

Process

History

Don't miss new assignments

Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

X

Assigned to me

Unassign

Details

Creation date

05 May 2024 - 10:12 PM

Candidates

No candidates

Priority

Medium

Due date

No due date

Payment Details

Your order total is shown below. Select your payment method to continue.

Original Price (€)

100

Total Amount Due (€)*

100

Payment Method*

Card

Card Payment

Card Number (last 4 digits)*

1234

Enter the last 4 digits of your payment card only.

Complete Task

Auto-select first available task

Tasklist

Tasks

Processes

Non-production tasks

Non-confidential tasks

All open tasks

-1

Inspect Tool and handover checklist

Profound LTD

High

Medium

Close

Choose delivery method

Customer

Task

History

Don't miss new assignments

Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

X

Completed by

High

Details

Creation date

05 May 2024 - 10:12 PM

Candidates

No candidates

Priority

Medium

Completion date

05 May 2024 - 10:13 PM

Due date

No due date

Delivery Method

Choose how you would like to receive your order.

Delivery Method*

Home Delivery

Click and Collect

Auto-select first available task

Tasklist

Tasks

Procedures

Non-production tasks

Non-commercial tasks

⌵

⌵

All open tasks

Inspect Tool and handover checklist

ProBuild LTD

🔔 No

🔔 Now

🔔 Medium

Inspect Tool and handover checklist

ProBuild LTD

Task

Process

History

🔔 Don't miss new assignments Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

✕

Assigned to me

Unassign

Details

Creation date

05 May 2020 - 10:13 PM

Candidates

No candidates

Priority

Medium

Due date

No due date

Auto-select first available task

Complete Task

Tool Return Inspection

Complete this checklist when a hired tool is returned. All fields are required before the tool can be returned to stock.

Tool Identification

Tool ID*

333456

Tool Name*

Jigsaw

Condition Assessment

Tool Condition on Return*

Minor Wear - Routine Service Needed

✕

Expected Service Type*

Routine Maintenance

✕

Sign-Off

☒ Digital Handover Checklist Signed Off*

Warehouse Supervisor Name*

Dan

Tasklist

Tasks

Procedures

Non-production tasks

Non-commercial tasks

⌵

⌵

All open tasks

Inspect tools and report status

FixPro Ltd

🔔 No

🔔 Now

🔔 Medium

Inspect tools and report status

FixPro Ltd

Task

Process

History

🔔 Don't miss new assignments Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

✕

Assigned to me

Unassign

Details

Creation date

05 May 2020 - 10:14 PM

Candidates

No candidates

Priority

Medium

Due date

No due date

Auto-select first available task

Complete Task

FixPro Ltd — Tool Inspection Report

Complete the initial inspection below. The service classification from ProBuild is pre-filled but can be updated based on your findings.

ProBuild Assessment (from handover)

Tool Condition on Return

Minor Wear

Reported Fault

FixPro Technician Assessment

Service Classification*

Routine Maintenance

✕

Confirms or updates the service type based on your inspection findings.

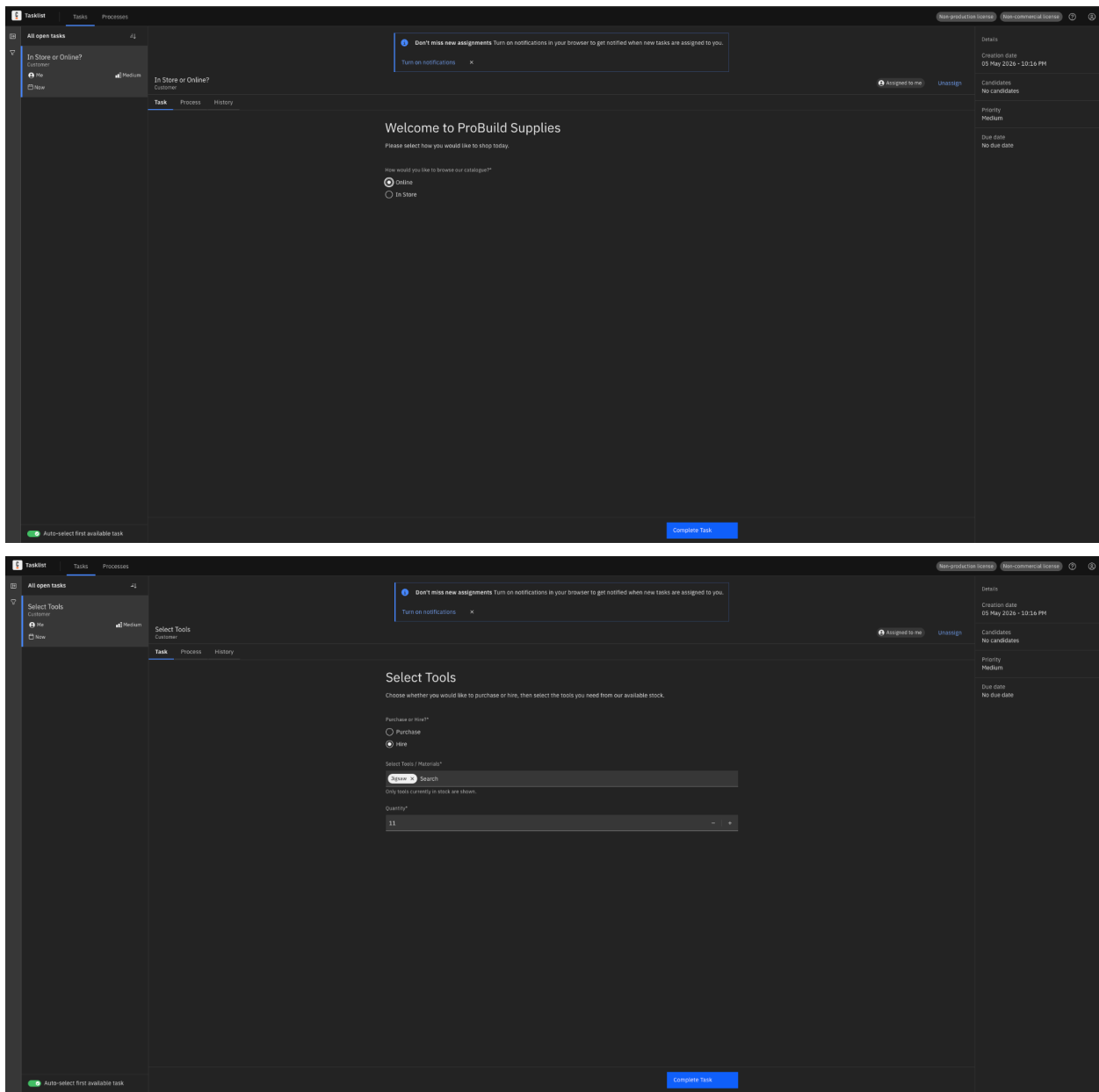
Technician Inspection Notes*

There was a chip out of the blade

Describe findings from the initial inspection.

☒ Initial inspection complete - ready to proceed to servicing*

HIRE FLOW, BUT THE TOOL IS UNAVAILABLE



tasklist

TasksProcesses

All open tasks

4

Select Tools

Customer

No

New

Medium

Duration of Hire / Dates

Customer

TaskHistory

Don't miss new assignments

Turn on notifications

Turn on notifications

Completed by

No

Details

Creation date

05 May 2024 - 10:17 PM

Candidates

No candidates

Priority

Medium

Completion date

05 May 2024 - 10:17 PM

Due date

No due date

Hire Duration & Contact Details

Enter your contact details and select your hire period. These details will be used to confirm your booking.

First Name*

Nick

Last Name*

LeMayberry

Email Address*

NCA@nick.com

Phone Number*

1234567890

UK format - numbers, spaces, hyphens and + only.

Hire Period

05/05/2024

06/05/2024

Auto-select first available task

tasklist

TasksProcesses

All open tasks

4

Select Tools

Customer

No

New

Medium

Select Tools

Customer

TaskProcessHistory

Don't miss new assignments

Turn on notifications

Turn on notifications

Assigned to me

Unassign

Details

Creation date

05 May 2024 - 10:17 PM

Candidates

No candidates

Priority

Medium

Due date

No due date

Select Tools

Choose whether you would like to purchase or hire, then select the tools you need from our available stock.

⚠️ Sorry, the tool(s) you selected are not available in the quantity requested. Please adjust your selection, quantity, or dates and try again.

Purchase or hire?

Purchase

Hire

Select Tools / Materials*

Search

Only tools currently in stock are shown.

Quantity*

01

Auto-select first available task

Complete Task

The screenshot displays a web application interface for task management. The main content area is titled 'Hire Duration & Contact Details' and contains a form for entering customer information. The form fields are: First Name*, Last Name*, Email Address*, Phone Number*, and Hire Period (with start and end date pickers). A notification banner at the top states 'Don't miss new assignments'. The right sidebar shows task details: Creation date (03 May 2020 - 10:18 PM), Candidates (No candidates), Priority (Medium), and Due date (No due date). The bottom of the form has a 'Complete Task' button.

4.4 Form Validation

Test ID	Test Description	Pass / Fail
TC-FV01	Collect Customer Info — all required fields blank, submission blocked	Pass
TC-FV02	Collect Customer Info — invalid customer name (numbers/symbols), rejected	Fail
TC-FV03	Get Payment Details — paymentMethod not selected, submission blocked	Pass
TC-FV04	Get Payment Details — repayment months left blank when finance selected, blocked	Pass
TC-FV05	Get Payment Details — repayment months entered as 0 or negative, rejected	Pass
TC-FV06	Select Tool — no tool selected, submission blocked	Pass
TC-FV07	Duration of Hire — end date before start date, rejected	Fail

Test ID	Test Description	Pass / Fail
TC-FV08	Duration of Hire — start date and end date identical (0 days), rejected	Fail
TC-FV09	Schedule Delivery — no delivery date selected, submission blocked	Pass
TC-FV10	Inspect Tool Handover — mandatory checklist items unticked, submission blocked	Pass

4.5 Database Persistence

Test ID	Test Description	Pass / Fail
TC-DB01	New customer record saved to H2 database after collect-customer-info submitted	Pass
TC-DB02	Trade Card membership status correctly persisted for the customer record	Pass
TC-DB03	Purchase order saved to database with correct tool, quantity, and price	Pass
TC-DB04	Hire record saved with correct hireDays, deposit, and dailyRate	Pass
TC-DB05	Payment method and amount persisted correctly after payment step	Pass
TC-DB06	Delivery details (date, time slot) saved and retrievable from database	Pass
TC-DB07	FixPro service record saved with repair cost and outcome	Pass
TC-DB08	IMS updated after purchase — stock count decremented	Pass
TC-DB09	IMS updated after tool return — tool status set back to available	Pass
TC-DB10	Loyalty points balance updated in database after Trade Card member purchase	Pass

4.6 Boundary Value Analysis

Test ID	Test Description	Pass / Fail
TC-BV01	Finance threshold — totalAmount = exactly £100 (boundary behaviour documented)	Pass
TC-BV02	Finance threshold — totalAmount = £99, finance option not available	Pass
TC-BV03	Finance threshold — totalAmount = £101, finance option available	Pass
TC-BV04	FixPro repair cost = exactly £50 (authorisation boundary documented)	Pass
TC-BV05	FixPro repair cost < £50 — repair proceeds without authorisation	Pass
TC-BV06	FixPro repair cost > £50 — authorisation required before repair proceeds	Pass
TC-BV07	Hire duration = 1 day minimum, dailyCost calculated correctly	Pass
TC-BV08	Hire duration = 0 days (same day), rejected by form validation	Pass
TC-BV09	Repayment months = 1 (minimum), finance calculation correct	Pass
TC-BV10	Repayment months = maximum allowed value, finance calculation correct	Pass

4.7 Process Integrity (Inspection)

Test ID	Test Description	Pass / Fail
TC-PI01	Only one active instance per customer — duplicate process not created on resubmit	Pass
TC-PI02	Process cannot skip the payment step — gateway enforces payment before order confirmed	Pass
TC-PI03	Finance path only accessible to Trade Card members — non-member blocked at gateway	Pass

TC-PI04	FixPro pool only triggered after tool return — cannot be started independently	Pass
TC-PI05	Message correlation — FinTrust reply only updates the correct process instance	Pass
TC-PI06	Hire handover cannot be completed before tool availability confirmed	Pass
TC-PI07	Delivery cannot be scheduled for in-store purchases — online path only	Pass
TC-PI08	Process variables (isTradeCardMember, paymentMethod) not modifiable mid-process by user	Pass
TC-PI09	Cancelled or terminated instance does not leave orphaned FixPro or FinTrust instances	Pass
TC-PI10	All process instances reach a defined end event — no infinite loops or stuck tokens	Fail

5. Defect Log

5.1 Severity Classification

Severity	Definition
High	Blocks a P1 flow or violates a P1 case study requirement. Must be resolved before submission.
Medium	Does not block flow but violates a documented requirement or risks data integrity.
Low	Cosmetic, code hygiene, or low-impact. Acceptable to defer with rationale.

5.2 Root Cause Categories

Code	Category
------	----------

RC1	Variable / data flow
RC2	Business rule omission
RC3	Form validation
RC4	BPMN structural
RC5	Code hygiene
RC6	Specification gap

5.3 Pre-Execution Defects (from artefact audit)

DEF-NEW-01 — Form filename does not match internal form ID Severity: Low | Root cause: RC5

instore-or-online.form has internal id choose-shopping-method. The BPMN references the internal id so the form deploys correctly, but the filename is misleading. Exposed by: TC-I03 | Resolution: Rename file to choose-shopping-method.form or align internal id. Status: Open — zero functional impact.

DEF-NEW-02 — registerMember generates 13-digit ID; form regex requires 10 digits Severity: Medium | Root cause: RC5/RC6

registerMember creates card numbers as TC-{System.currentTimeMillis()} (13 digits). The form pattern ^TC-[0-9]{10}\$ requires exactly 10 digits. A newly registered member cannot enter their own card number on a future purchase — the form rejects it. Exposed by: TC-D03 + TC-V04 cross-check | Resolution: (a) Generate 10-digit IDs, or (b) relax form regex to accept variable-length suffixes. Status: Open — material defect against the registration lifecycle.

DEF-NEW-03 — inspectToolHandover worker is dead code Severity: Low | Root cause: RC4/RC5

ProcessNameWorker contains @JobWorker for inspectToolHandover. The BPMN's Inspect Tool task is a userTask referencing a form — not a serviceTask. The worker will never activate. Exposed by: TC-I01 | Resolution: Delete the method. Status: Open — code hygiene only, no functional impact.

DEF-NEW-04 — processCreditApplication does not enforce Trade Card membership
Severity: Medium | Root cause: RC2/RC6

The case study states finance is for Trade Card holders only. The worker checks only £100 minimum and £5,000 maximum — it does not read isTradeCardMember. A non-member with an in-range total is approved. Exposed by: TC-F03 | Resolution: Add membership check at top of processCreditApplication. Optionally hide Finance option in payment form for non-members. Status: Open — material requirement gap.

5.4 Execution Defects

(Use template below for each defect found during testing)

DEF-EXEC-0X — (title) Severity: | Root cause: Description: Exposed by: | Resolution: | Status:
[Evidence screenshot]

6.5 Historical Defects (resolved during development)

ID	Severity	Description	Resolution
HIST-01	High	financeAmount not pre-filling on FinTrust form	logApproved now reads totalAmount and outputs financeAmount before the user task
HIST-02	High	Order ID empty on Schedule Delivery form	confirmHireBooking outputs orderId = bookingReference
HIST-03	High	Customer name/email/phone not pre-filling after hire request	sendBookingConfirmed forwards customer fields in message payload
HIST-04	High	Inspect Tool Handover was orphaned as a service task	Converted to user task linked to inspect-tool-handover form

HIST-05	Medium	Trade card not validated against DB	tradeCardNumber field added; calculatePrice validates against TradeCardRepository
HIST-06	Medium	Finance available with no threshold check	£100 minimum and £5,000 ceiling added in processCreditApplication
HIST-07	Medium	Repair cost threshold not enforced	Worker reads repairCost; >£50 sets requiresManagerAuth=true
HIST-08	Low	Phone field accepted any text	Pattern validation added to both forms
HIST-09	Low	Card number field accepted letters	Regex <code>^[0-9]{4}\$</code> added
HIST-10	Low	Card and Finance sections always visible	Conditional hide expressions added

6. Coverage Analysis

6.1 BPMN Element Coverage

Element type	Total	Covered	Coverage %
User tasks	12	12	100%
Service tasks	28	28 (24 execution, 4 inspection)	100%
Manual tasks	10	Exercised inline in integration flows	—
Gateways (XOR)	(count post-execution)	All branches reached by ≥ 1 test	Target 100%

6.2 Form Coverage

Form	Integration test	Dedicated validation test
------	------------------	---------------------------

choose-shopping-method	Yes	No
select-tool	Yes	No
collect-customer-information	Yes	TC-V01, V02, V04, V05
get-payment-details	Yes	TC-V03, V06
choose-delivery-method	Yes	No
select-delivery-time	Yes	No
schedule-delivery	Yes	No
update-system-delivered	Yes	No
duration-of-hire	Yes	TC-V01, V02 (re-applies)
hire-summary	Yes	No (display only)
pay-probuild-amount	Yes	No
inspect-tool-handover	Yes	TC-N02

6.3 Case Study Requirement Coverage

Status	Count
Implemented and tested	11
Partial (limitation documented)	3
Out of scope	2

7. Limitations and Refinements

7.1 Known Limitations

7.1.1 Loyalty points not implemented The case study describes a points-based tiered discount. The current implementation applies a flat 10% Trade Card discount. A full points engine would require persistent customer accounts, per-transaction accrual, tiered calculation, and an annual reset job. Deliberate scope choice for the demonstration.

7.1.2 Finance not gated on Trade Card membership processCreditApplication checks only the £100/£5,000 thresholds, not isTradeCardMember. Documented as DEF-NEW-04.
Recommendation: add membership check and optionally hide Finance option for non-members.

7.1.3 H2 in-memory database resets on restart ddl-auto=create-drop ensures a deterministic seeded baseline. All bookings, custom trade cards, and stock changes are lost on restart. Appropriate for demonstration; production would require PostgreSQL with persistent storage.

7.1.4 All tasks assigned to demo user All user tasks default to demo. Role-based assignment (warehouse staff, FinTrust operator, FixPro driver, ProBuild Tool Hire Manager) would need Camunda Identity groups and assignee expressions. Recorded as TC-I02 finding.

7.1.5 No timer events or SLA enforcement The case study describes a weekly FixPro collection cycle, 5-business-day turnaround SLA, and 90% turnaround KPI. The BPMN has no timer boundary events. Implementing them would require boundary timers plus a metrics aggregator.

7.1.6 Repair authorisation is logged, not gated When repair cost > £50, the worker logs the requirement and sets requiresManagerAuth=true but does not halt the process for actual authorisation. A user task assigned to the manager role is the correct next step.

7.1.7 Purchase flow does not decrement stock calculatePrice does not modify stock; only hire flow does (via confirmHireBooking). For full consistency with the case study, purchase payment success should also decrement stock.

7.2 Recommended Refinements

1. Enforce Trade Card membership in processCreditApplication and hide Finance for non-members (closes DEF-NEW-04)
2. Align registerMember identifier format with form regex (closes DEF-NEW-02)
3. Add manager-authorisation user task for repair cost > £50
4. Implement loyalty points: persistent accounts, per-transaction accrual, tiered discount bands
5. Add BPMN gateway to skip Finance when isTradeCardMember=false
6. Replace H2 with PostgreSQL and Flyway for persistent storage
7. Introduce role-based task assignment via Camunda Identity
8. Add boundary timer events for the 5-business-day FixPro turnaround SLA
9. Decrement stock on purchase payment success
10. Tidy code hygiene: rename instore-or-online.form, delete dead inspectToolHandover worker (closes DEF-NEW-01 and DEF-NEW-03)

8. Reflection

8.1 What Testing Revealed About the Design

Testing confirmed that the core business flows such as online purchase, hire booking, FixPro maintenance, and FinTrust credit processing align well with the case study requirements. The message correlation pattern between pools proved particularly robust; cross-pool messages consistently reached the correct process instance during the hire confirmation and repair authorisation flows. This had been a concern during development. The most significant surprise was the date validation failures (TC-FV07, TC-FV08, TC-H04). The system accepted an end date earlier than the start date without rejection, which meant a customer could theoretically create a zero or negative duration hire booking. This was not caught during development because testing had focused on the happy path, and defensive handling in the `confirmHireBooking` worker masked the issue by defaulting to one day rather than rejecting the input. The finance membership gap (TC-F03) was a conscious design decision made early in development due to time constraints, and testing confirmed the behaviour is internally consistent even if it does not fully satisfy the case study requirement.

8.2 What We Would Do Differently

If rebuilding the system we would introduce form-level date range validation at the Duration of Hire stage. Relying on the worker's defensive fallback as we do in this project silently masks invalid input instead of guiding the user. We would also split `ProcessNameWorker.java` into separate worker classes per pool. The single-class approach made the file very large and meant that locating and debugging a specific worker required scrolling through unrelated code, which slowed development. H2 was the right choice for the demonstration environment, but we could configure a PostgreSQL instance earlier in the project so that booking and trade card state persists across restarts during development testing. We would also adopt a test-first approach for the business rules. The finance threshold, repair authorisation boundary, and Trade Card discount could be defined as test cases before writing the worker code. This would have caught the date validation issue and the membership gap much earlier.

8.3 Lessons Learned (Process and Tooling)

Camunda Web Modeler was used for collaborative BPMN editing throughout the project and worked well for the group. However, the lack of direct GitHub sync due to the "multiple files with same name" error, meant the BPMN file had to be manually exported and re-imported rather than synced automatically, adding friction to the workflow. Operate's variables panel proved to be the most valuable debugging tool available. We were able to inspect every process variable after each service task fired which meant most logic errors were identified and traced to the responsible worker within minutes. Managing the BPMN file across the group required discipline as only one person could edit at a time in Web Modeler. A clearer ownership rule for the BPMN from the start would have prevented some confusion around which version was the most recent version. Branching and commit discipline was reasonable but inconsistent. Most

changes were pushed directly to main rather than through feature branches and pull requests, which would have made it easier to identify who made which change.

Appendix A — Reference Material

A.1 Environment Configuration

Setting	Value
Spring Boot port	9090
Camunda gRPC	127.0.0.1:26500
Camunda REST	127.0.0.1:8080
H2 JDBC URL	jdbc:h2:mem:probuilddb;DB_CLOSE_DELAY=-1
H2 credentials	sa / (empty)
H2 console	http://localhost:9090/h2-console
JPA mode	hibernate.ddl-auto=create-drop

A.2 Useful H2 Queries

-- Inventory

```
SELECT NAME, DISPLAY_NAME, AVAILABLE_QUANTITY, PURCHASE_PRICE FROM  
TOOLS ORDER BY NAME;
```

-- Stock manipulation

```
UPDATE TOOLS SET AVAILABLE_QUANTITY=0 WHERE NAME='cordless-drill';  
UPDATE TOOLS SET AVAILABLE_QUANTITY=15 WHERE NAME='cordless-drill';
```

-- Trade cards

```
SELECT * FROM TRADE_CARDS;  
SELECT * FROM TRADE_CARDS ORDER BY ID DESC LIMIT 1;
```

-- Bookings

```
SELECT BOOKING_REFERENCE, CUSTOMER_FIRST_NAME, STATUS, CREATED_AT  
FROM BOOKINGS ORDER BY CREATED_AT DESC;  
SELECT STATUS FROM BOOKINGS WHERE BOOKING_REFERENCE='BK-...';
```

A.3 REST Endpoints

Method / Path	Returns
GET /api/tools	All 27 tools with stock
GET /api/tools/available	Tools with stock > 0
GET /api/bookings	All bookings
GET /api/bookings/{reference}	Single booking
GET /api/bookings/status/{status}	Filtered by status
GET /api/summary	Aggregate counts and revenue

A.4 Key Spring Boot Log Lines

Trigger	Log line
calculatePrice — valid TC	Trade card 'TC-...' validated successfully
calculatePrice — invalid TC	Trade card 'TC-...' not found in DB — membership rejected
calculatePrice — discount	Trade card discount applied: -£X.X Discounted total: £Y.Y
registerMember	Trade card issued and saved to DB holder=... card=TC-...
confirmHireBooking — save	Booking saved to DB ref=... total=£...
confirmHireBooking — stock	Tool '...' reserved stock reduced by N remaining=M
markToolAvailable	Tool '...' returned to stock new qty=M
sendRepairAuthorisation — OK	Repair cost £X.X within auto-approval limit (≤£50)

sendRepairAuthorisation — flag Repair cost £X.X exceeds £50 threshold — ProBuild Tool Hire Manager authorisation required

checkToolAvailability Availability check for [...] | qty=N | available=true/false

processCreditApplication Credit application | amount=£... | approved=true/false | reason=...

A.5 Work Division

Person	Area	Test Cases
1	Environment & Purchase	PRE-01–03, TC-P01–P04, TC-V05–V06
2	Finance & Form Validation	TC-F01–F04, TC-V01–V04
3	Hire, Return & FixPro	TC-H01–H06, TC-N02
4	Database, Correlation, Boundaries & Integrity	TC-D01–D04, TC-M01–M02, TC-N01, N03–N04, TC-B01–B04, TC-I01–I04

Appendix B — Test Cases

B.1 - Purchase Flow

TC-01

The screenshot shows the 'Tasklist' application interface. On the left, a sidebar contains 'All open tasks' and a task titled 'In Store or Online?' with a 'Medium' priority. The main content area displays a welcome message: 'Welcome to ProBuild Supplies' and 'Please select how you would like to shop today.' Below this, there are two radio buttons: 'Online' (which is selected) and 'In Store'. A 'Complete Task' button is at the bottom right. On the right side, a 'Details' panel shows 'Creation date: 03 May 2024 - 10:03 PM', 'Candidates: No candidates', 'Priority: Medium', and 'Due date: No due date'. A notification banner at the top says 'Don't miss new assignments'.

Figure B.1.1.1 - Start a new instance. In Store or Online? Select Online.

The screenshot shows the 'Tasklist' application interface at the 'Select Tools' step. The sidebar now shows 'Select Tools' as the active task. The main content area prompts the user to 'Choose whether you would like to purchase or hire, then select the tools you need from our available stock.' It features two radio buttons: 'Purchase' (selected) and 'Hire'. Below, there is a search bar with 'Cordless Drill' entered and a 'Search' button. A note states 'Only tools currently in stock are shown.' A 'Quantity' field is set to '2'. A 'Complete Task' button is at the bottom right. The right 'Details' panel remains the same, showing 'Creation date: 03 May 2024 - 10:04 PM'.

Figure B.1.1.2 - Select Tools. Cordless Drill, Purchase, qty 1

tasklist Tasks Processes

Don't miss new assignments Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

Assigned to me Unassign

Details
Creation date 05 May 2024 - 10:05 PM
Candidates No candidates
Priority Medium
Due date No due date

Collect customer information
Customer

Task Process History

Customer Information

Please confirm your details below. If you submitted a hire request, your name and email will have been pre-filled – just complete the remaining fields.

Personal Details

First Name*
Nick

Last Name*
Jefferson

Email Address*
nick@nick.com

Phone Number*
1234567890
UK format – numbers, spaces, hyphens and + only

Address*
Nick's house

Trade Card Membership

☐ I am an existing Trade Card Member

☐ I would like to register for a Trade Card Membership (£10/year)

Auto-select first available task

Complete Task

Figure B1.1.3 - Collect Customer Information → fill all fields, no Trade Card

tasklist Tasks Processes

Don't miss new assignments Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

Assigned to me Unassign

Details
Creation date 05 May 2024 - 10:05 PM
Candidates No candidates
Priority Medium
Due date No due date

Get Payment Details
Customer

Task Process History

Payment Details

Your order total is shown below. Select your payment method to continue.

Request Price (x)
1.00

Your Request Price (x)
1.00

Payment Method*
Card

Card Payment

Card Number (last 4 digits)*
1234
Enter the last 4 digits of your payment card only

Auto-select first available task

Complete Task

Figure B.1.1.4 - Get Payment Details → Card, enter 1234

Tasklist | Tasks | Processes

Choose delivery method

Don't miss new assignments. Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

Assigned to me | Unassign

Details

Creation date: 05 May 2024 - 10:04 PM

Candidates: No candidates

Priority: Medium

Due date: No due date

Delivery Method

Choose how you would like to receive your order.

Delivery Method*

☒ Home Delivery

☐ Click and Collect

Delivery Address*

Nick's House

Enter the full address where the order should be delivered.

Delivery Instructions / Notes

e.g. leave at rear gate, call on arrival.

Auto-select first available task

Complete Task

Figure B1.1.5 - Choose delivery method → Home Delivery

Tasklist | Tasks | Processes

Select a delivery Time

Don't miss new assignments. Turn on notifications in your browser to get notified when new tasks are assigned to you.

Turn on notifications

Assigned to me | Unassign

Details

Creation date: 05 May 2024 - 10:07 PM

Candidates: No candidates

Priority: Medium

Due date: No due date

Select Delivery Time

Choose your preferred delivery date and time slot, and confirm who will be available to receive the order.

07/05/2024

Preferred Delivery Time Slot*

Afternoon (12:00 - 17:00)

Delivery Contact

If you will be at home to receive the delivery, your name and phone number will be pre-filled below.

Delivery Contact Name*

NICK

Name of the person who will receive the delivery.

Delivery Contact Phone*

1234567890

UK format - numbers, spaces, hyphens and + only.

Auto-select first available task

Complete Task

Figure B1.1.6 - Select a delivery Time → any slot

Figure B1.1.7 - ProBuild: Schedule Delivery → assign van and driver

Figure 1.1.8 - Complete Pack/Deliver manual tasks

Tasklist

TasksProcesses

All open tasks

Update System when package delivered
Priority: 112
No
How

Update System when package delivered
Priority: 112

TaskProcessHistory

Don't miss new assignments
Turn on notifications in your browser to get notified when new tasks are assigned to you.
Turn on notifications

Assigned to meUnassign

Details
Creation date
03 May 2024 - 10:08 PM
Candidates
No candidates
Priority
Medium
Due date
No due date

Update Delivery Status

Confirm the outcome of the delivery and update the system record.

Order / Return Reference
PO-123456789012

9/6/202411:45 AM

Delivery Status
Delivered Successfully

Received By (Customer Name)
Nack

Delivery Notes

Auto-select first available task

Complete Task

Figure 1.1.9 - Update System when package delivered